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Prediction of European Winter 2023/24: Influence of a Strong El Niño

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ABSTRACT

UK winter 2023/24 had a particularly stormy start, similar to 2015/16, another strong El Niño winter. Successfully predicted by the Met Office seasonal forecast system, this increased early-winter storminess was associated with lower-than-average surface pressure to the west of the UK. Although late winter during El Niño is often associated with the negative phase of the North Atlantic Oscillation (NAO), in 2023/24 the NAO was near-neutral in observations and weakly positive in the ensemble mean forecast, the late-winter Atlantic surface pressure pattern in the forecast resembling the response observed during the very strongest El Niño events. Both observations and the forecast showed relatively higher pressure west of the UK, but north-west Europe and the southern North Atlantic were more cyclonic than forecast. The forecast ensemble mean stratospheric polar vortex (SPV) was relatively strong, and unusually, the mean of ensemble members which had a sudden stratospheric warming (SSW) did not project strongly onto the negative NAO, analysis suggesting tropospheric teleconnections masked the usual SSW impact. Conversely, the observed SPV was weak; however, a mid-January SSW had little surface impact. Overall, our analysis suggests that, despite some differences in late winter, El Niño was the principal driver of observed conditions in the North Atlantic-European sector.

1 | Introduction

This paper describes the Met Office seasonal forecast and its evaluation for winter (DJF) 2023/24. It forms part of a series of case studies examining real-time winter forecasts, with the aim of fostering a better understanding of the influences of the various drivers that give rise to predictive skill for the UK and wider European region. Although internal atmospheric variability plays a large role in the mid-latitudes, potentially useful skill has been demonstrated for a variety of sectors for the UK (e.g., Thornton et al. 2019; Stringer et al. 2020). As a key driver of skilful forecasts, attention is often focused on the North Atlantic Oscillation (NAO) (Scaife et al. 2014; Weisheimer et al. 2017; Stringer et al. 2024) although, more recently skill has also been demonstrated for the East Atlantic (EA) pattern (Molteni and Brookshaw 2023; Thornton et al. 2023; O'Reilly 2025). However, the ability of seasonal forecast models to forecast these modes of North Atlantic variability differs from year to year (Baker et al. 2024), and signals in both have been shown to be too weak, being subject to the signal-to-noise paradox (Dunstone et al. 2016; Scaife and Smith 2018). The East Atlantic/Western Russia and Scandinavia teleconnection patterns can also have a significant influence on European climate (Barnston and Livezey 1987), however, analysis indicates both had relatively weak amplitude during winter 2023/24 (NOAA/CPC 2024).

A number of factors had the potential to influence European climate anomalies during winter 2023/24. The Quasi-Biennial

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Oscillation had entered an easterly phase which tends to favour a weakened stratospheric polar vortex (SPV) and negative NAO (Gray et al. 2018). Closer to Europe, there were large positive sea surface temperature (SST) anomalies in the tropical Atlantic, which would project onto the negative North Atlantic tripole pattern (Sutton et al. 2000; Rodwell and Folland 2002), again favouring a negative NAO. Anomalously low Arctic sea-ice would also favour a negative NAO (Dunstone et al. 2016; Chen, Kumar, et al. 2024a); however, impacts may not be large (Smith et al. 2022) and while below average, anomalies were not extreme (Copernicus 2024). The Indian Ocean dipole (IOD) was, however, in a positive phase, which has been shown to be associated with a positive NAO (Hardiman et al. 2020; Abid et al. 2021). In addition, recent research has shown that the La Niña observed in the previous winter 2022/23 (Stringer et al. 2024) might also favour the negative phase of the NAO in this subsequent winter (Scaife et al. 2024). Perhaps most significantly, the backdrop to this forecast was a strong and well predicted El Niño event in the tropical Pacific whose possible impacts (e.g., Toniazzo and Scaife 2006; Ayarzagüena et al. 2018) were a major consideration in creating the winter outlook.

The outline for this paper is as follows: Section 2 details the current Met Office seasonal prediction system, methods and data. Section 3 describes the winter 2023/24 outlook and evaluation and Section 4 discusses the key drivers of predictability. A summary and conclusions are given in Section 5.

2 | Method

The Met Office seasonal forecast system GloSea6 (GS6) (Kettleborough et al. 2026) was used to make the winter forecast. Two forecast members are initialised each day, and the dates used for the 2023/24 winter outlook from November range from 30th October to 19th November 2023, giving a lagged ensemble of 42 members. Hindcast sets, each with a 7-member ensemble, cover 24 winters from 1993/94 to 2016/17. Hindcasts initialised on 1st, 9th and 17th November (504 members) are used to calculate the model climatology and forecast anomalies. These dates lie within the forecast initialisation period, with the aim of minimising any initialisation date dependence. The underpinning climate model in GS6 is based on the Met Office Unified Model (UM) version GC3.1 (Williams et al. 2018) which has a vertical resolution of 85 levels in the atmosphere (top at 85 km) and 75 levels in the ocean. The ocean horizontal resolution is 0.25° on a tri-polar grid and the atmosphere has a horizontal resolution of 0.83° (longitude) × 0.56° (latitude).

Data used for verification are from HadUK (Hollis et al. 2019), HadISST (Rayner et al. 2003) and ERA5 (Hersbach et al. 2020). Daily means of 10 hPa zonal mean wind are calculated from 00, 06, 12 and 18Z ERA5 values. A major sudden stratospheric warming is defined to occur when the 10 hPa zonal mean wind at 60° N becomes easterly; details follow the algorithm of Charlton and Polvani (2007). The categorization of El Niño events follows Toniazzo and Scaife (2006) using area average Niño3 (5° N-5° S, 150 W-90° W) SST anomalies with 'strong' events defined as occurring when Niño3 > 1.5°C, and 'moderate' events 0.83 < Niño3 ≤ 1.5°C. The NAO index is calculated as the difference in sea level pressure (PMSL) between two boxes located around Iceland (25°–16° W, 63°–70° N) and the Azores (28°–20° W, 36°–40° N). Anomalies for ERA5 and the model forecast are calculated with respect to the model hindcast period, 1993–2016, the climatological period used for HadUK is 1991–2020.

3 | Results

3.1 | The Met Office Winter Forecast

The forecast ensemble mean PMSL anomaly for DJF (Figure 1a) shows a somewhat bland signal over the North Atlantic/European sector, with a weak positive NAO of 2.4 hPa. However, the monthly forecast pressure charts show the possibility of rather different circulation types in early and late winter. For December, low pressure is positioned to the north-west of the UK, with relatively higher pressure to the south (Figure 1b), suggesting a pattern that is associated with bringing relatively mild, wet and windy weather from the Atlantic, with an increase in storminess. January and February are broadly similar (the average is shown here) and build high pressure over and to the south-west of the UK (Figure 1c). The late winter pressure pattern indicates the potential for more settled conditions developing at times and there is an increased chance of north-westerlies and colder weather relative to earlier in the winter.

The Met Office 3-month outlook for winter (DJF) 2023/24 (Figure 2a), which was issued to U.K. Government departments and the public in November 2023, indicated moderately increased chances of mild, wet and windy conditions for the season as a whole (see Lockwood et al. 2023 for more details of the recently developed wind/storminess forecasts). The accompanying issued text highlighted the likely intraseasonal shift:

- *For the winter as a whole, mild conditions are a little more likely than cold conditions.*

- *Greatest chance of impacts from cold weather are in late winter.*

- *There is a slightly increased chance of wet conditions and impacts from heavy rainfall, especially in early winter.*

- *Stormy conditions and impacts from high winds are a little more likely than normal, especially in early winter.*

The possible impacts from rain and wind were of particular concern given the preceding wet autumn (Sandford et al. 2024).

3.2 | Evaluation

The observed NAO for DJF 2023/24 was −1.0 hPa (Figure 1d). This is very weak considering the interannual standard deviation of the DJF NAO is about 8 hPa and is also therefore in reasonable agreement with the 2.4 hPa weak positive NAO indicated by the GS6 ensemble mean forecast (Figure 1a), with the observed value lying within the ensemble forecast spread. Verifying observations of anomalies of mean temperature, accumulated rainfall and mean wind speed from HadUK (Hollis et al. 2019) were 3.7°C (+3.7σ, standard deviations),

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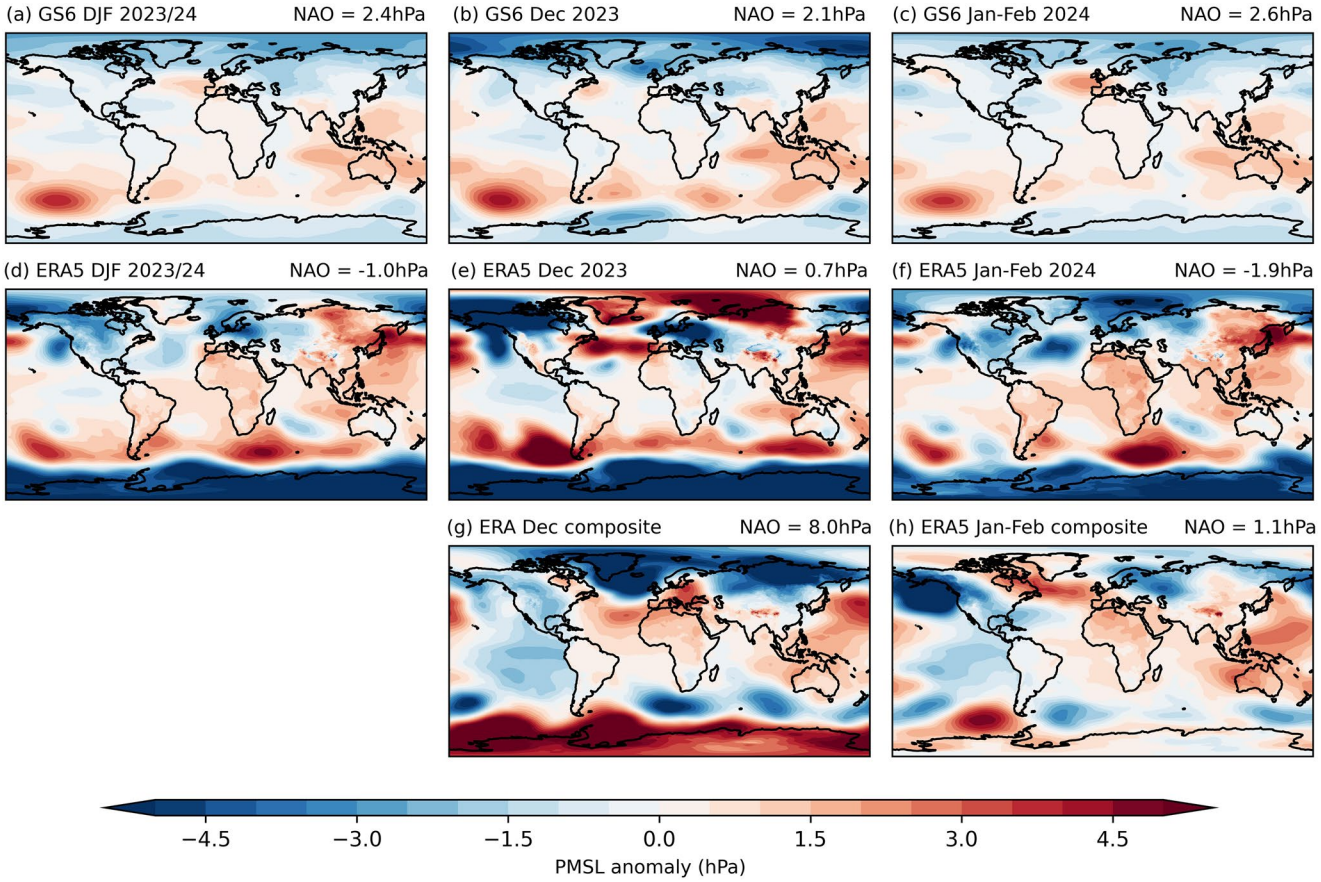


FIGURE 1 | Winter PMSL patterns. Winter 2023/24 PMSL anomalies for: GS6 ensemble mean prediction for (a) DJF, (b) December and (c) January–February; Observations (ERA5) for (d) DJF, (e) December, and (f) January–February; Composite over recent 'strong' El Niño events (Niño3 > 1.5°C), 1972/73, 1982/83, 1997/98, 2015/16, for (g) December and (h) January–February.

100.9 mm (+3.4σ), and 1.4 mph (+1σ) respectively (Figure 2b). These values lie in the highest quintile for temperature and rainfall. Overall, wind speed was in the near normal category but was higher than average during December. These values agree well with the issued forecast, coinciding with the raised probability of wet, mild and windy conditions for the winter as a whole.

In December, the observed low pressure in the Atlantic was to the southeast of the forecast position, extending eastward across Europe (Figure 1e), and was a continuation of an unusually strong and persistent positive EA pattern which had been present since October (Rantanen et al. 2024; NOAA/CPC 2024). The north–south dipole in the vicinity of the UK showed a similar pattern but with a stronger meridional gradient than the signal from the ensemble mean forecast (Figure 1b), and this period saw the UK affected by a succession of Atlantic depressions, with 7 named storms occurring between the end of October and early January (Table 1). The pattern of observed PMSL anomalies for January–February (Figure 1f) also somewhat resembles the forecast (Figure 1c) with low pressure in the southern North Atlantic, relatively higher pressure extending from south of Greenland to Spain, and low pressure centred to the north-east of the UK over the Norwegian/Barents Sea. However, details of the positioning and strength of the anomalies clearly differ; the observed southern North Atlantic low pressure is deeper and centred in the central Atlantic to the north-east of the forecast low pressure, the region of relatively higher pressure is further north and considerably weaker, and the low pressure to the north-east is deeper and more extensive. For a period in January, high pressure dominated and the weather was calm and settled; however, late January saw two more named storms. In February, the cyclonic area extended further south than suggested by the ensemble mean forecast (Figure 1c), although this aspect was captured by some individual ensemble members. This resulted in very mild and very wet weather, especially for southern regions of England, with more depressions tracking in from the southwest. February 2024 was the warmest February on record for England and Wales, and southern England had the wettest February on record (NCIC 2024).

4 | Remote Influences

Slowly varying patterns in SST are an important driver of climate anomalies and 2023 had been a year of unusual warmth in the global ocean (Johnson and Lumpkin 2024). Prominent features include the El Niño event in the tropical Pacific which peaked in winter 2023/24, a warm tropical Atlantic, and the positive phase of the IOD which, although in decline, still persisted in the Indian Ocean (Figure 3b). For the most part, these anomalies were very well forecast by GS6 (Figure 3a), although the tropical Atlantic anomalies are somewhat weaker than observed.

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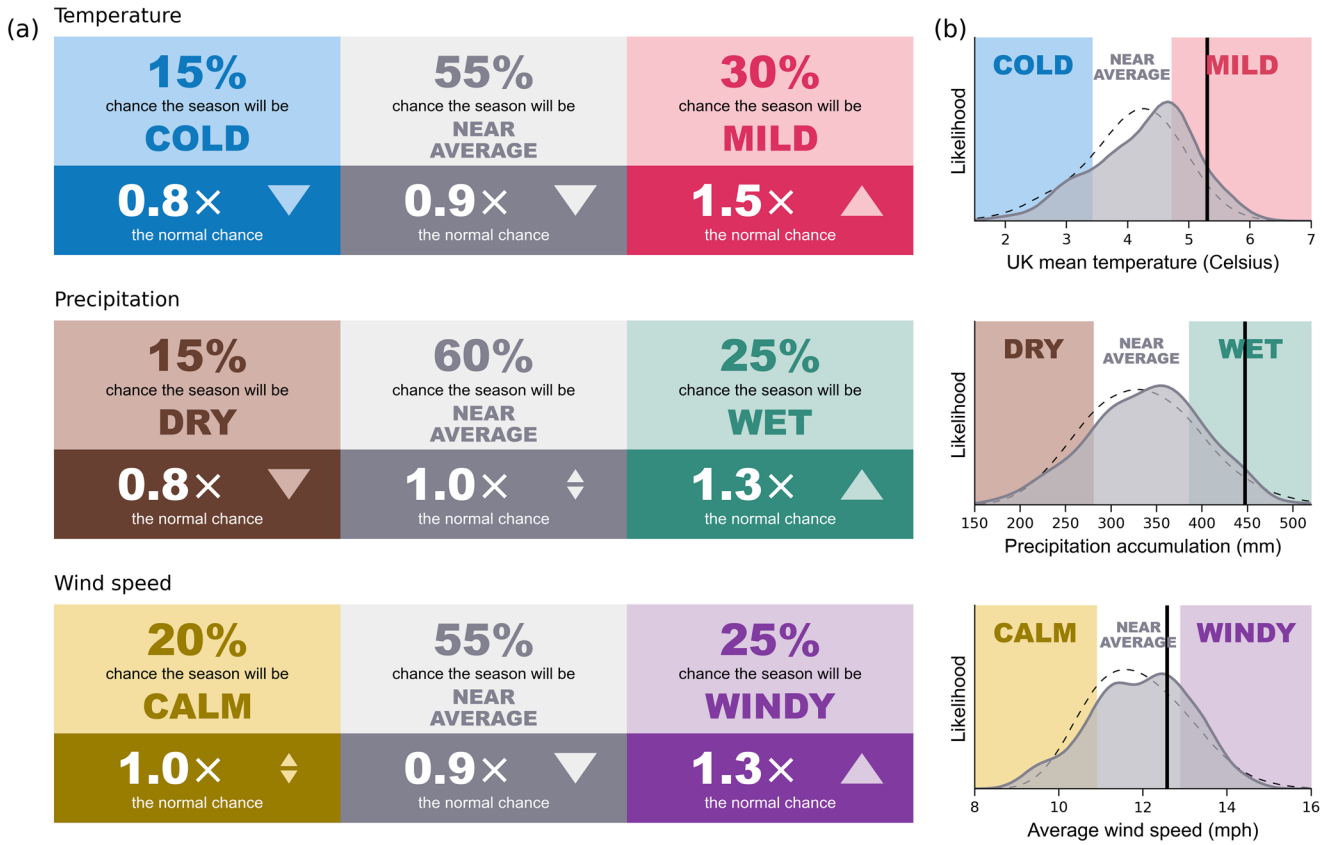


FIGURE 2 | Met Office winter outlook 2023/24. Issued UK-mean December–January–February (DJF) seasonal outlook and verification: (a) Quintile seasonal forecast for DJF 2023/24 for UK-mean temperature, precipitation, and windspeed issued at the end of November 2023. The percentages refer to the likelihood of the seasonal outcome occurring in the lowest (left boxes), middle three (middle boxes), and highest (right boxes) quintiles. The climatological occurrences of these are 20%, 60%, and 20% respectively. The factor change in the climatological occurrence is indicated by the numbers at the bottom of each box. (b) The climatological (dashed lines) and forecast (solid lines with grey shading) distributions for each variable. The observed outcome is shown by the solid black vertical lines.

TABLE 1 | Named storms in the UK, October 2023 to March 2024.

Name	Date of impact on UK and/or Ireland and/or Netherlands
Babet	18–21 October 2023
Ciarán	1–2 November 2023
Debi	13 November 2023
Elin	9 December 2023
Fergus	10 December 2023
Gerrit	27–28 December 2023
Henk	2 January 2024
Isha	21–22 January 2024
Jocelyn	23–24 January 2024

Note: Data from https://weather.metoffice.gov.uk/warnings-and-advice/uk-storm-centre/uk-storm-season-2023-24.

From early in 2023 the Met Office forecast system had been indicating the likely development of El Niño (https://www.metoffice.gov.uk/research/climate/seasonal-to-decadal/gpc-outlooks/el-nino-lanina), and, by early summer, there were clear indications of a strong event developing for winter 2023/24. From the UK perspective, a focus for interpretation of winter forecasts was the likely magnitude of the event. Historically, El Niño winters over the North Atlantic/European region show contrasting conditions through the winter, with an intraseasonal shift from positive to negative phase NAO conditions occurring between the earlier (November–December) and later (January–February) months (Moron and Gouirand 2003; Ayarzagüena et al. 2018). However, the very strongest El Niño events (Niño3 > 1.5°C) tend to have a rather different late winter response over the Atlantic, with positive pressure anomalies over the East Atlantic extending to the UK and southwestern Europe, with low pressure anomalies to the north-east and south-west, thought to be associated with a Rossby wave train emanating from the tropical Atlantic (Toniazzo and Scaife 2006).

A timeseries of Niño3 anomalies for recent El Niño events which peak in the 'moderate' and 'strong' categories are shown in Figure 3c. Also shown are the observed and GS6 forecast anomalies for 2023/24. We see that the ensemble mean forecast agreed well with observations, and although this El Niño event was weaker in DJF than the extreme events of 2015/16, 1997/98 and 1982/83 (Huang et al. 2016), at 1.7°C it was similar in magnitude to 1972/73, which also showed a 'strong' El Niño response over the North Atlantic/European sector (see Toniazzo and Scaife, their Figure 2).

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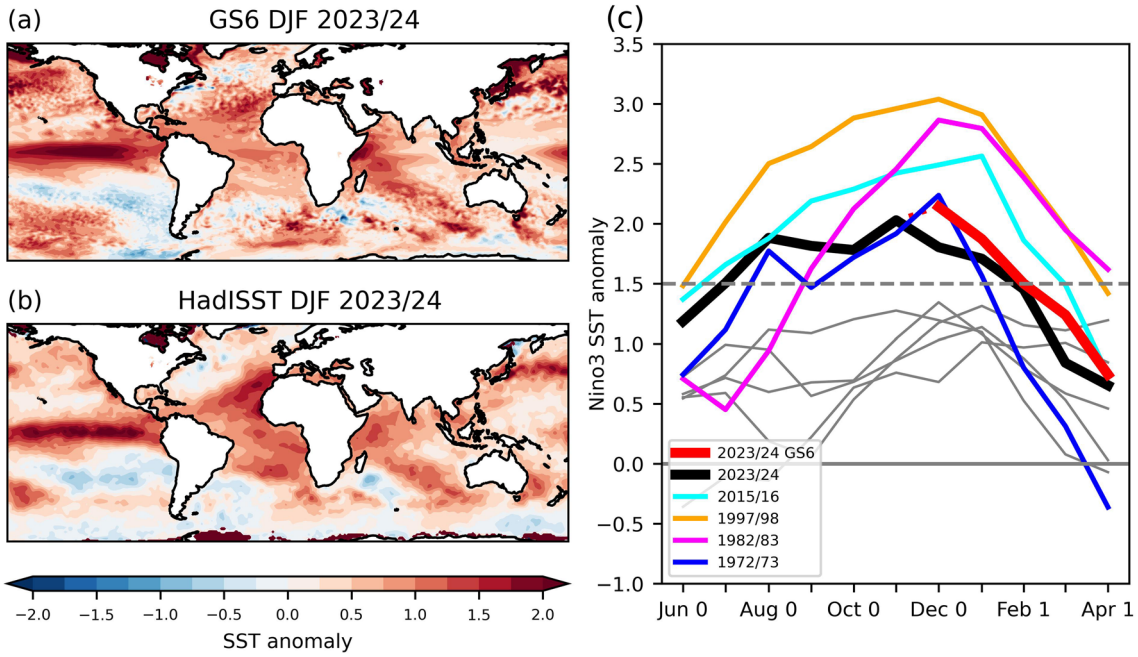


FIGURE 3 | Winter sea surface temperature patterns. Global DJF SST anomaly for (a) GS6 ensemble mean forecast, and (b) observations (HadISST). (c) Timeseries of Niño3 (5°N-5°S, 150W-90°W) SST anomalies from June (prior to peak amplitude) to April (post peak) for observed strong (colours) and moderate (grey) El Niño events from 1950 onwards. Observations for 2023/24 are shown in black, and the ensemble mean GS6 forecast initialised in November is shown in red. The dashed grey line at 1.5°C indicates the boundary between 'moderate' and 'strong' El Niño.

The observed composite PMSL response for the four most recent strong El Niño events for December and January–February (Figure 1g,h) is in reasonable agreement over the Atlantic with the ensemble mean GS6 2023/24 forecast (Figure 1b,c). As discussed above, the observed anomalies for December (Figure 1e) look a little different for the canonical EA pattern, and in January–February (Figure 1f) while there is broad agreement in the patterns, details of the positioning and strength of the anomalies differ.

Further evidence supporting a possible stationary Rossby wave train emanating from the tropical Atlantic is shown by the 200 hPa stream function, with the forecast (Figure 4a), observations (Figure 4b) and strong El Niño composite (Figure 4c) all showing a generally similar pattern, with a wave train consisting of a cyclonic (node 1), anticyclonic (node 2) and cyclonic (node 3) centres (and likewise in the southern hemisphere). These patterns equate closely with the respective PMSL anomalies (Figures 1c,f,h), suggesting the features are equivalent barotropic and that details in the positioning of the nodes are closely associated with our weather regimes. Interestingly, we also see in the observations, and to a lesser extent in the forecast, an area of anomalous Rossby wave source extending across the entire width of the Atlantic. Both this and the enhanced anticyclonic region (node 2) were similarly seen in 1998 (Figure 4d), although the positioning is somewhat further north in 1998. This was another strong El Niño year which also had a negative Atlantic SST tripole pattern, suggesting a possible role for the warm Atlantic tropical SST anomalies in modulating the teleconnection, although the relationship between tropical Atlantic SST anomalies and the NAO has been shown to undergo multidecadal variability (Chen et al. 2015).

Next, we consider the role of the SPV. On occasions, the normally westerly winter SPV undergoes a significant collapse. Although usually relatively short lived, these events, known as major sudden stratospheric warmings (SSWs), can have surface impacts which can last for up to 2 months and are characterised by the negative phase of the NAO (Baldwin and Dunkerton 2001).

The likelihood of an SSW occurring has been shown to increase in El Niño winters relative to ENSO-neutral years (Butler and Polvani 2011; Ineson et al. 2024) and this feature is reproduced in seasonal hindcasts (Scaife et al. 2016). However, for winter 2023/24 the GS6 forecast indicated that the risk of an SSW occurring was less than the average chance in the hindcast climatology (Figure 5a) and that the SPV was likely to be somewhat stronger than average throughout the winter season (Figure 5b), although a proportion of individual forecast members do have SSWs from mid-December onwards and by the end of the DJF season about 45% (19 members) have had an SSW.

In contrast to our forecast, observations show that while the SPV was initially strong during November, in the following months there were frequent disturbances. From mid-November onwards the vortex gradually weakened, culminating in a mid-winter SSW on 16th January. Following this there was a recovery; however, and then a further event that reduced the zonal mean zonal wind to 0.4 ms^{-1} on 19th February. A second, more intense and persistent SSW occurred on 15th March. A detailed analysis of the SPV behaviour in 2023/24 can be found in Lee et al. (2024), and Rao et al. (2024).

SSWs can have a major impact on weather and climate in the North Atlantic/European area with cooler and drier conditions

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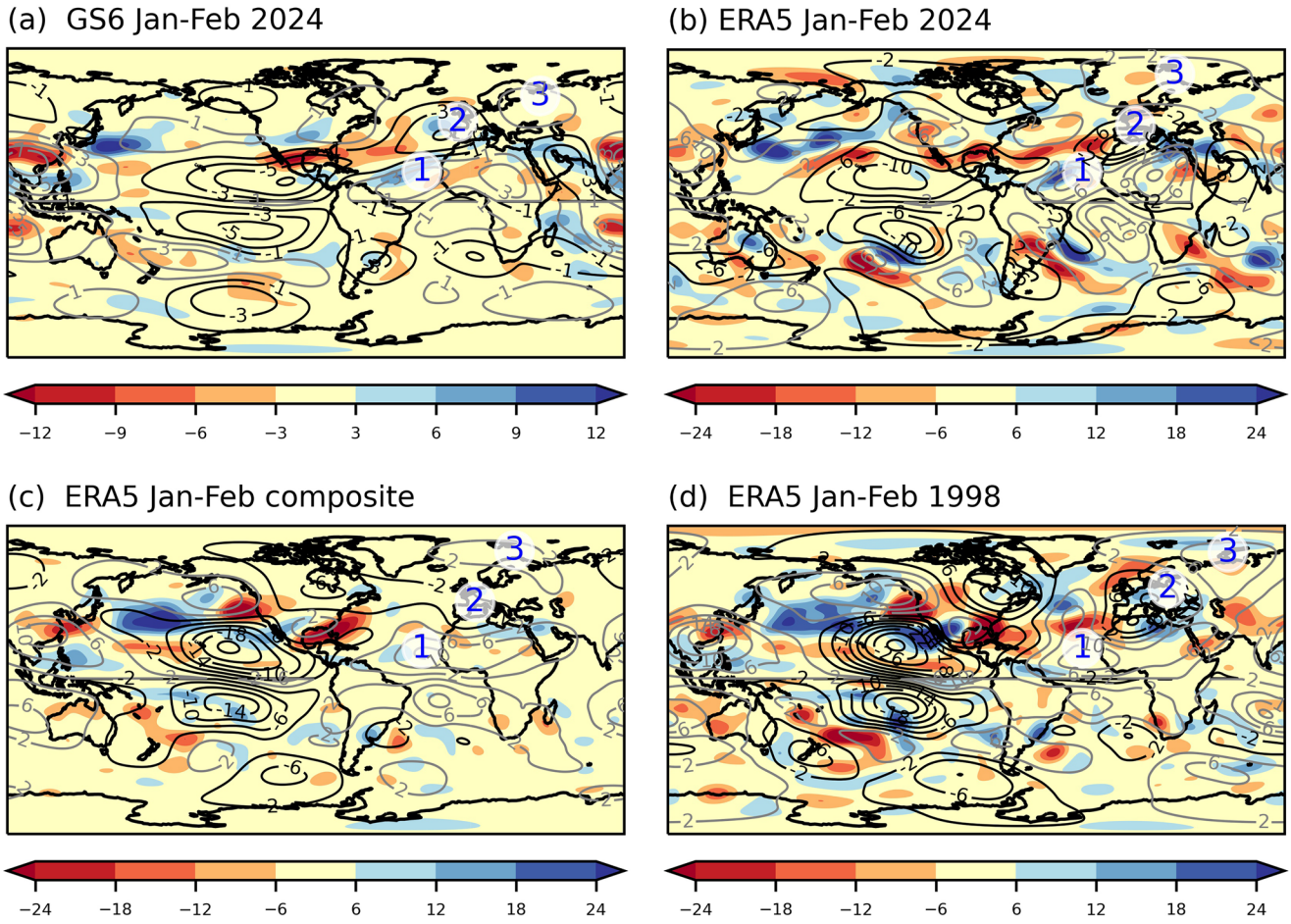


FIGURE 4 | Upper-level circulation anomalies. 200 hPa stream function (zonal mean removed) (black/grey contours represent anticyclonic/cyclonic circulation, km^2s^{-1}) and Rossby wave source diagnostics (colours, $10^{-11}\,s^{-2}$) for January–February for (a) GS6 ensemble mean forecast 2024; Observations (ERA5) for (b) 2024 (c) composite over recent 'strong' El Niño events 1973, 1983, 1998, 2016, and (d) 1998. Numbers (blue) indicate suggested Rossby wave path (compare with Toniazzo and Scaife 2006—their Figure 4).

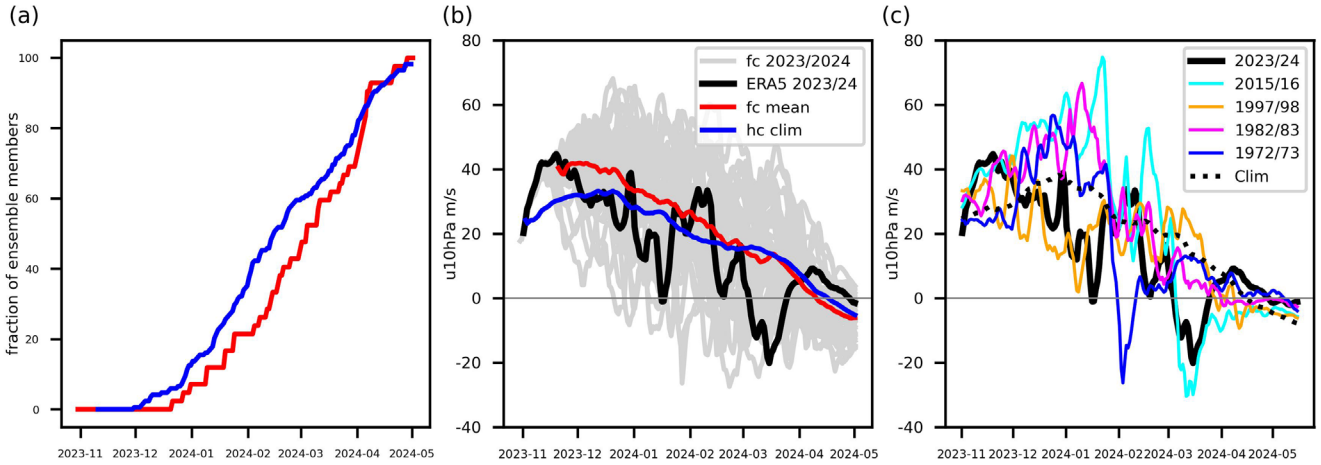


FIGURE 5 | Polar vortex strength. Timeseries of (a) The cumulative fraction of GS6 ensemble forecast members that have experienced a major SSW event (red) compared with the hindcast members (blue), (b) observed (ERA5) 60° N 10 hPa zonal mean wind (black), with GS6 forecast ensemble members (grey), the GS6 forecast mean (red) and GS6 climatology (blue), (c) observed (ERA5) 60° N 10 hPa zonal mean wind during years with previous recent strong El Niño events (colour) and ERA5 climatology (dashed).

for northern Europe (Baldwin et al. 2021 and references therein). However, the observed January–February PMSL for 2024 (Figure 1f) suggests that on average, the mid-January SSW had very little, if any, significant surface impact, with the average NAO for this period near zero (–1.9 hPa). We consider the SPV strength in other recent strong El Niño winters. SSWs did not

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occur in the most recent 3 strong El Niño events, although we see that 1997/98 had a rather weak SPV with a minor warming, almost meeting the SSW criterion on early January (Figure 5c). In 1972/73 there was an SSW at the end of January. We note that the January–February PMSL anomaly patterns for all these events (not shown), while differing somewhat in detail, all displayed a wave-like response in the tropical Atlantic (Figure 1h) rather than the negative phase of the NAO.

As we have seen, the GS6 ensemble mean forecast for winter 2023/24 (Figure 1a) contains a variety of SPV solutions (Figure 5b). Compositing the January–February PMSL forecast into cases with SSWs (Figure 6a) and without SSWs (Figure 6b) shows that for this winter the mean SSW composite does not project strongly onto the negative NAO, although individual members (not shown) do vary. We see that the observed response in 2023/24 (Figure 1f) bears some similarities to the SSW composite, with lower pressure over on the eastern seaboard of the U.S. and low pressure over Europe, though shifted southward in comparison with observations. The observed response however does not replicate the high pressure anomaly at high latitudes seen in the SSW composite. Differencing the SSW and non-SSW composites (not shown) gives a pattern closely resembling the negative NAO, suggesting that, at least for GS6, the usual SSW influence is simply being masked by other tropospheric teleconnections.

5 | Conclusions

The Met Office winter outlook for 2023/24 gave good guidance, with increased chances of early winter storminess, the enhanced chances of mild and wet conditions, and an intraseasonal shift in these conditions correctly anticipated. However, some details were not well captured by the ensemble mean alone, with February 2024 being more cyclonic than anticipated. Over the extended winter season (October–March) there were 9 named storms, which was second only to the 11 named storms in 2015/16 (Met Office, named storms launched in 2015), another strong El Niño year. The storminess and associated high rainfall caused significant disruption in terms of flooding, transport disruption and power outages (Kew et al. 2024). The rainfall also had significant impacts on agriculture with damage to planted crops and difficulties establishing new crops due to waterlogged ground (Poynting and Vladev 2024). Overall, the DJF NAO in observations was near neutral. As the NAO is a primary driver of skill in the seasonal forecast system, winter 2023/24 provides an interesting case study where observed pressure patterns projected only very weakly onto the NAO (see also Knight et al. 2017; Dunstone et al. 2018).

Our analysis suggests a key driver of the North Atlantic/European climate anomalies in winter 2023/24 was the well forecast, strong El Niño event, with the forecast ensemble mean teleconnection patterns largely resembling the patterns from past strong events. In particular, there are some similarities in upper-level circulation with the 1997/98 El Nino event, an event which coincided with a negative North Atlantic tripole pattern, and similar warm SST anomalies in the tropical Atlantic. However, there are also differences in detail in the response, demonstrating the complexity of the combinations of various drivers (e.g., Senan et al. 2024), or the role of unpredictable variability. While in general, large-scale observed pattern correlation has been found to increase with the observed strength of ENSO events (L'Heureux, Harnos, et al. 2024a), winter 2023/24 provides another case study of a strong event (see also Scaife et al. 2017) adding to evidence of a non-linear response to El Niño over the European/Atlantic sector (Toniazzo and Scaife 2006; Hardiman et al. 2019).

Although our forecast suggested a less-than-average chance of a major disruption to the SPV, an SSW did occur in mid-January and the SPV was observed to be disturbed and weak. In roughly 1/3 of observed SSW events, surface impacts fail to materialise (Bett et al. 2023). Interestingly, we see that on average, the model ensemble members that did have an SSW showed only a weakly negative NAO, suggesting that, at least in the model, other teleconnection patterns associated with the strong El Niño tended to dominate the Atlantic/European response. An alternative possibility lies in the details of this particular event; the analysis by Lee et al. (2024) indicates this may have been an example of a 'reflecting' SSW, and therefore would not have been expected to lead to long-lived disruption of the SPV and associated substantial negative NAO. Although the focus of this study has been on DJF, we note the overall contrast between this winter and the moderate El Niño winter of 2009/10, which similarly had two SSWs and one of the most negative observed NAOs on record (Fereday et al. 2012).

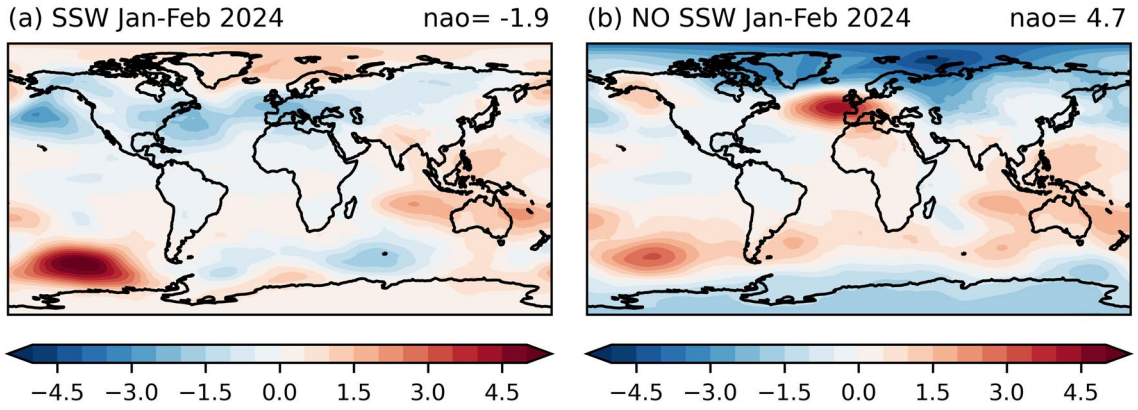


FIGURE 6 | Impact of an SSW in the winter forecast. GS6 ensemble mean forecast for January–February for (a) ensemble members which have an SSW prior to mid-February (31%), (b) ensemble members which do not have an SSW (69%).

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The reason for the relatively strong forecast SPV has not been investigated. However, we note that the forecast initialisation was during a period when the SPV was relatively strong compared with climatology (Figure 5c), possibly associated with teleconnections from the positive IOD (Hardiman et al. 2020; Abid et al. 2021), and that for December initialisation dates the ensemble mean SPV forecast became weaker (not shown). For this November initialisation, the forecast shows only a weak deepening of pressure in the Aleutian low area so that wave drag on the SPV may have been less in the model than might have been expected during an El Niño year (e.g., Ineson and Scaife 2009). This weakness would be consistent with the generally weak ENSO teleconnections found in seasonal prediction systems (Williams et al. 2023), however, observations were also somewhat muted in this respect, with a pattern showing a resemblance to that associated with years when the QBO is in an easterly phase (Garfinkel and Hartmann 2010). Analysis of the SPV has suggested that the initial vortex weakening at least was unrelated to tropospheric forcing (Lee et al. 2024).

We have presented forecast anomalies calculated relative to the GS6 hindcast period, 1993–2016. Recent studies are indicating that accounting for the climate change trend can play an increasing role in the interpretation of results (van Oldenborgh et al. 2021; Chen, Chen, et al. 2024b; Ciasto and Butler 2024; L'Heureux, Tippett, et al. 2024b). As such, achieving a balance between having enough years and ensemble members to provide a stable climatology from which to calculate forecast anomalies but also to be representative of current climate may also suggest a need for larger ensembles (Kettleborough et al. 2026).

Author Contributions

Sarah Ineson: conceptualization, investigation, formal analysis, writing – original draft, writing – review and editing, validation, visualization, methodology. **Chris Almond:** validation, data curation. **Paul Davies:** validation, data curation. **Nick J. Dunstone:** conceptualization, methodology, investigation, formal analysis, writing – review and editing, validation, software, supervision. **Chris K. Folland:** writing – review and editing. **Margaret Gordon:** data curation, software, validation, visualization. **Steven C. Hardiman:** writing – review and editing. **Jeff R. Knight:** writing – review and editing, data curation, validation. **Peter McLean:** software, validation, data curation, visualization. **Yu Nie:** writing – review and editing. **Adam A. Scaife:** writing – review and editing, investigation, validation, funding acquisition, supervision, conceptualization, methodology, formal analysis. **Doug M. Smith:** writing – review and editing. **Nicky Stringer:** writing – review and editing, visualization, data curation, methodology, validation. **Hazel E. Thornton:** writing – review and editing. **Brent Walker:** data curation, validation.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The GloSea6 and ERA5 data used in the paper are available from the C3S Climate data store https://cds.climate.copernicus.eu/#!/home. The Met Office Hadley Centre observations datasets, HadISST and HadUK, can be downloaded from https://www.metoffice.gov.uk/hadobs/.

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